# WEEK 01

# Introduction to Machine Learning (CS4613)

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# Rules

- You are here to learn.
- Do not create disturbance in the class.
- Good ethics will have a positive impact on improving overall grades.
- Time management is mandatory to at least pass the course.
- Asking questions about the relevant topic is much appreciated.
- Taking help from AI is appreciated, but as an assistant.



Capital University of Science & Technology
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# About Course

- Title: Machine Learning (ML)
- Code: CS4613
- Pre-req: CS3811 – Artificial Intelligence
- Credit Hours: 3 (2 hours for theory, 1 for practical implementation)
- Reference Book: Machine Learning: A Probabilistic Perspective by Kevin P. Murphy, 2012.
- Reference Link: https://www.tutorialspoint.com/machine_learning/index.htm
- Link: https://www.geeksforgeeks.org/machine-learning/machine-learning-tutorial/
- Purpose of this course:
  - Present the basic machine learning concepts
  - Present a range of machine learning algorithms along with their strengths and weaknesses
  - Enable students to apply machine learning algorithms to solve problems of moderate complexity

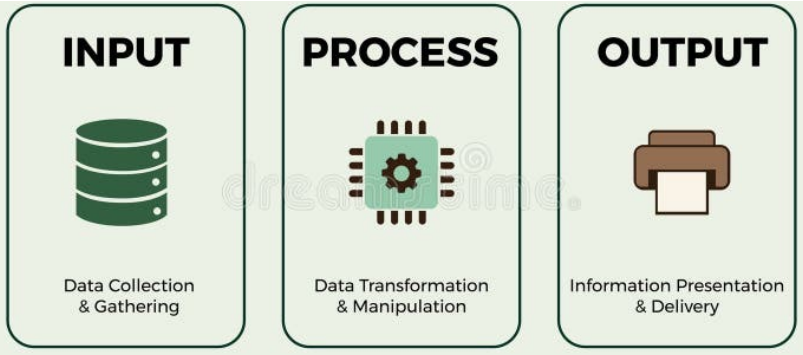
# Grading Policy

- The following is intended Overall Grading Policy of this course

| S. No. | Assessment Items | Percentage |
|---|---|---|
| 1 | Assignments | 10% |
| 2 | Quizzes | 10% |
| 3 | Semester Project/Presentation | 20% |
| 4 | Midterm Exam | 20% |
| 5 | Final Exam | 40% |

# Recall

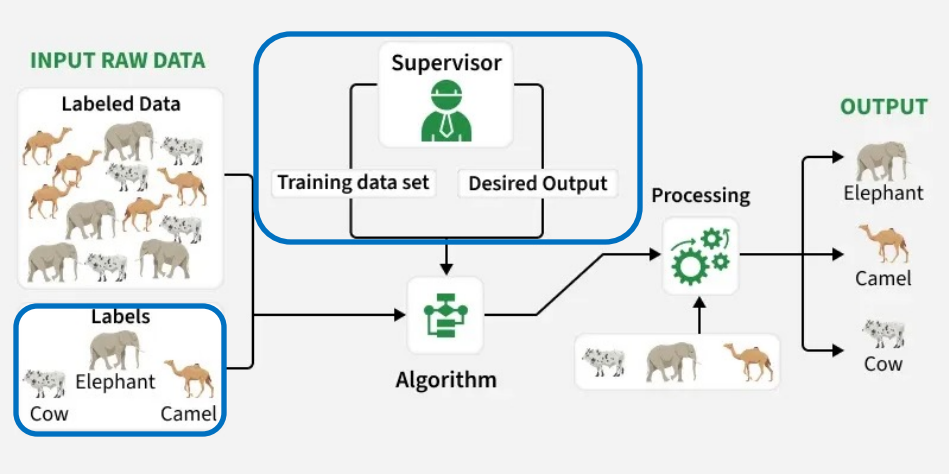
- *What is intelligence?*
- What to do, how to do, when to do, and why to do?
- **Cross road** on **zebra crossing** when **signal is red** to **reach other side**.
- *What is artificial intelligence?*
- When **machines** start **performing** the same **tasks** based on given **rules**.
- *Why AI?*
- Humans are biased and lazy in complex
- Machines are unbiased and perpetual.




- *What is a machine?*
- A human-made **device** that helps us **perform** a **task**.
- A calculator takes numbers, calculates, and gives an answer.
- *Why machine learning instead of AI?*
- Machines **learn** from **previous examples (data)** instead of fixed **rules**.

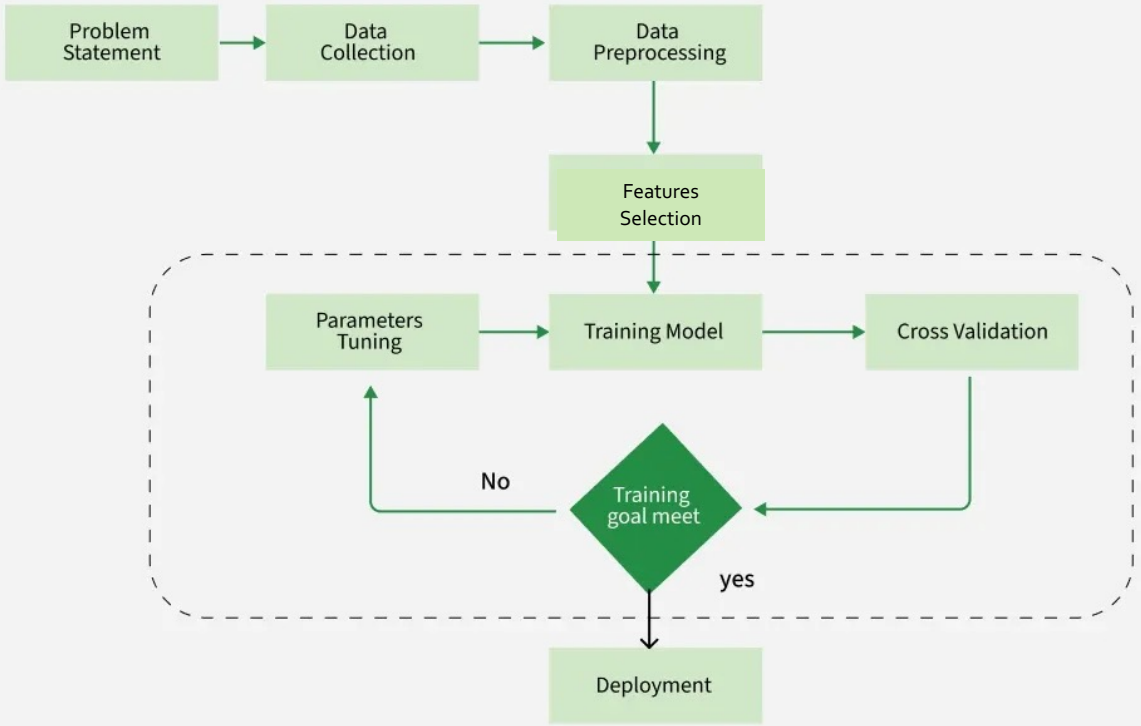
# Learning from Data

- What is ML?
- **Learns** from **data** by finding **patterns** and fixed **rules** and **improves** over **time**.
- What is not in AI?
- The Supervisor and the Labels
- Four main aspects of ML:
  - Experience: Labeled animals
  - Task: Identify new animal
  - Process: Algorithm steps
  - Performance: How correctly identified
- Formal Definition:
  - The ability of a machine to improve its performance on a specific task over time by learning patterns from experience.



# Flow of ML Model

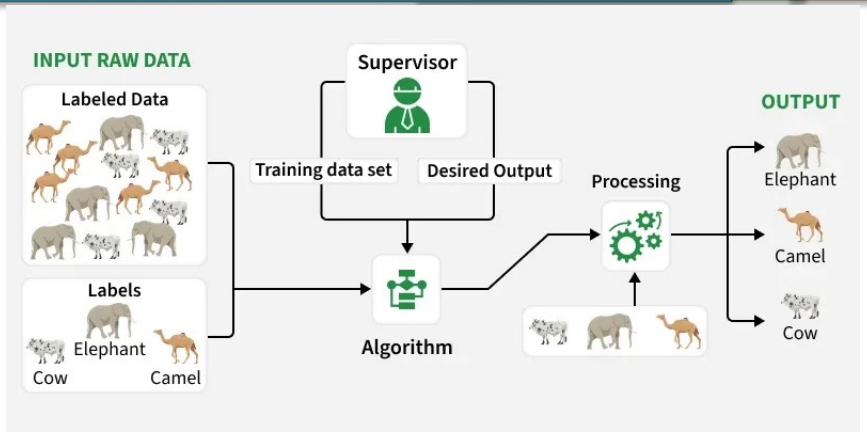
| Step | Example |
|---|---|
| **Problem statement** | "Predict the selling price of a house." |
| **Data collection** | Collect house size, location, bedrooms, age, and sale price. |
| **Features Selection** | Select the most relevant and informative data to be given to a model |
| **Choose model** | Type of algorithm. |
| **Train model** | It learns how size, location, and bedrooms relate to price. |
| **Cross-validation** | Check whether predicted prices are close to actual prices for houses not used in training. |
| **Training goal met?** | Is the average prediction error below, for example, PKR 500,000? |
| **Parameter tuning** | Change the tree depth or other model settings, then train again. |
| **Deployment** | Add it to a property website where users enter house details and receive an estimated price. |

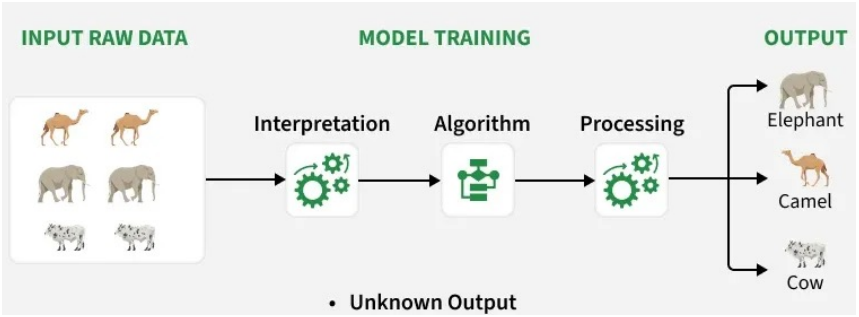


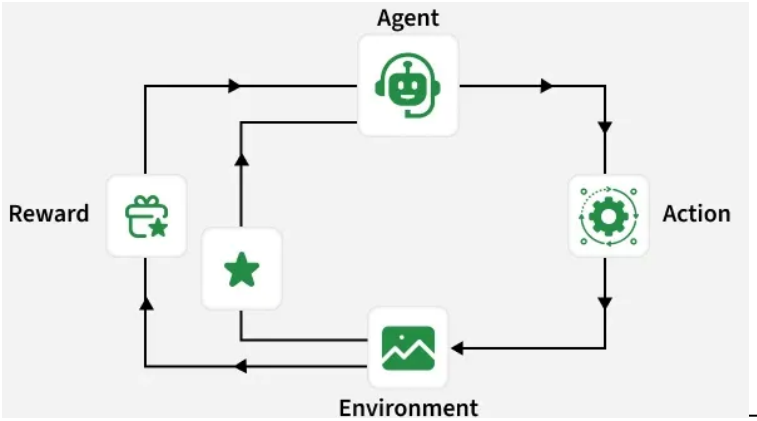

- Have you tried the example at home?
- Python code demo to implement the example
- input: 2000, 3000, 3, 2, 12, 8
- Predicted Output: ?

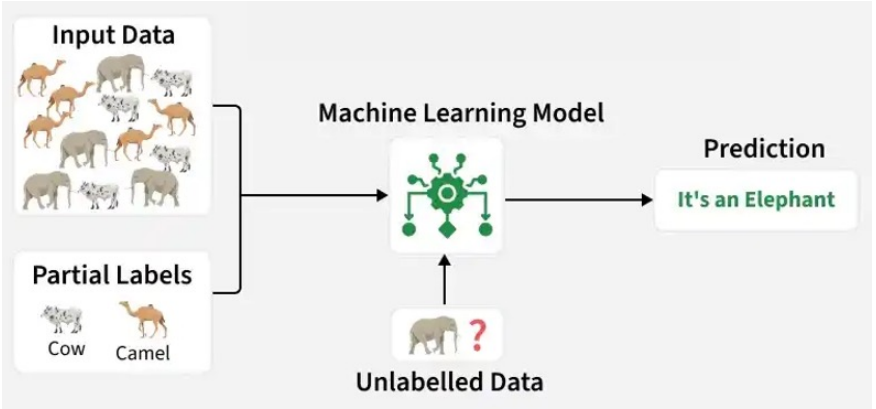
# Types of ML Methods

- Supervised Machine Learning
- Unsupervised Machine Learning
- Semi-supervised Machine Learning
- Reinforcement Machine Learning



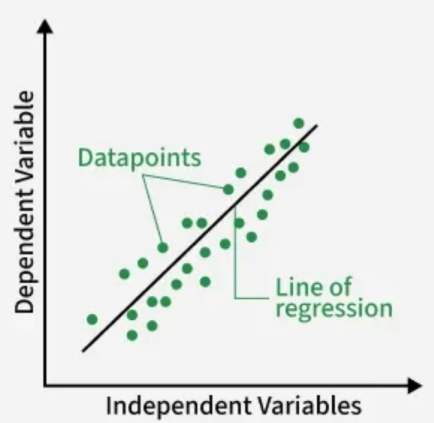



# Supervised ML

- Classification: A supervised ML technique that predicts labels from input data using extracted features.
- Predicts a category or label
- Example: car or not?



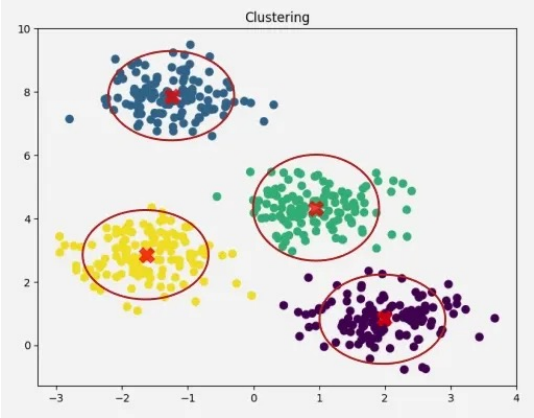

- Regression: Predicts continuous numerical values by learning relationships between input variables and an output variable.
- Predicts a number (continuous value).
- Example: Price of a house



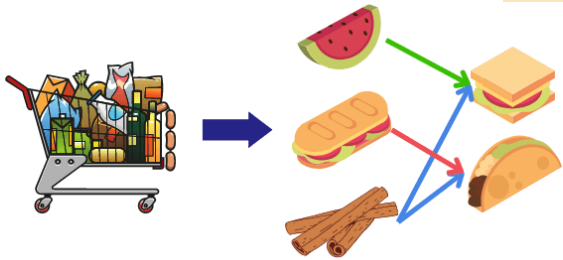



# Unsupervised ML

- Clustering: Groups similar data points without labeled data.
- Discover hidden patterns in similar data points.
- Example: Grouping different shapes (circle, square, triangle)

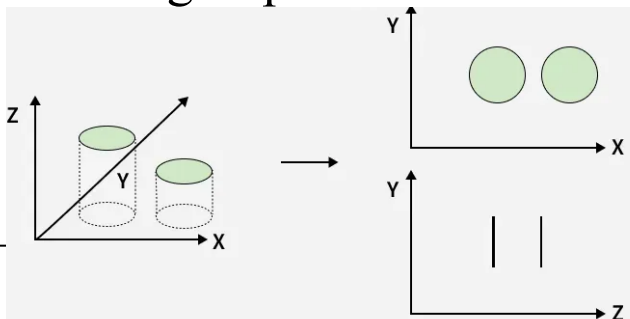



- Association Rules Mining:
- Find relationships, correlations, or patterns within large sets of data items.
- Describe how often items occur together.
- Example: {Bread, Butter}→{Milk}



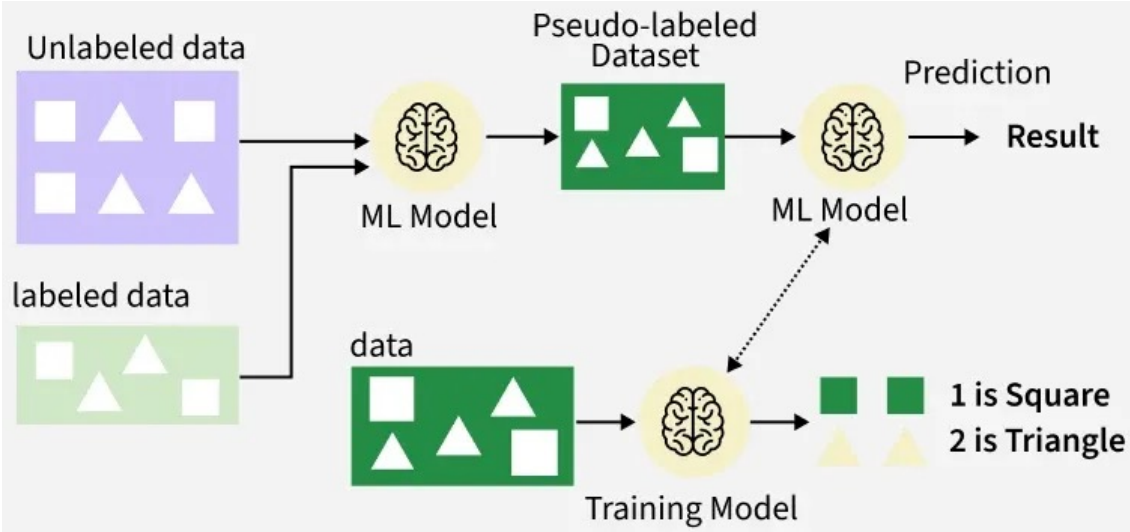
*"93% of people who purchased item A also purchased item B"*

- Dimensionality Reduction:
- Reduces the number of features in data while preserving important information.
- Transforms high-dimensional data into lower.
- Example: Feature selection



# Semi-Supervised ML

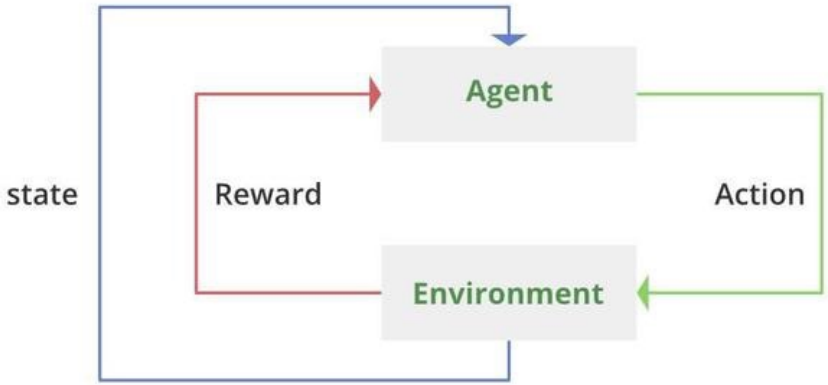
- Small amount of labeled data along with a large amount of unlabeled data.
- Learn a function that accurately predicts outputs based on inputs.




- Usage:
  - Labeled data is costly, e.g., medical imaging data.
  - Large volumes of labeled data, e.g., social media posts.
  - Manual labeling is time taking.
- Advantages:
  - Cost effective, handling unseen data effectively
- Chances of mistakes are high

# Reinforcement Learning

- Focuses on how agents can learn to make decisions through feedback.




- Flow:
  - Environment gives State to Agent
  - Agent chooses Action
  - Environment changes
  - Environment gives Reward and New State
  - Agent learns and repeats
- Solves complex sequential decision problems
- Learns from real-time interaction
- Computationally expensive


Thank You!